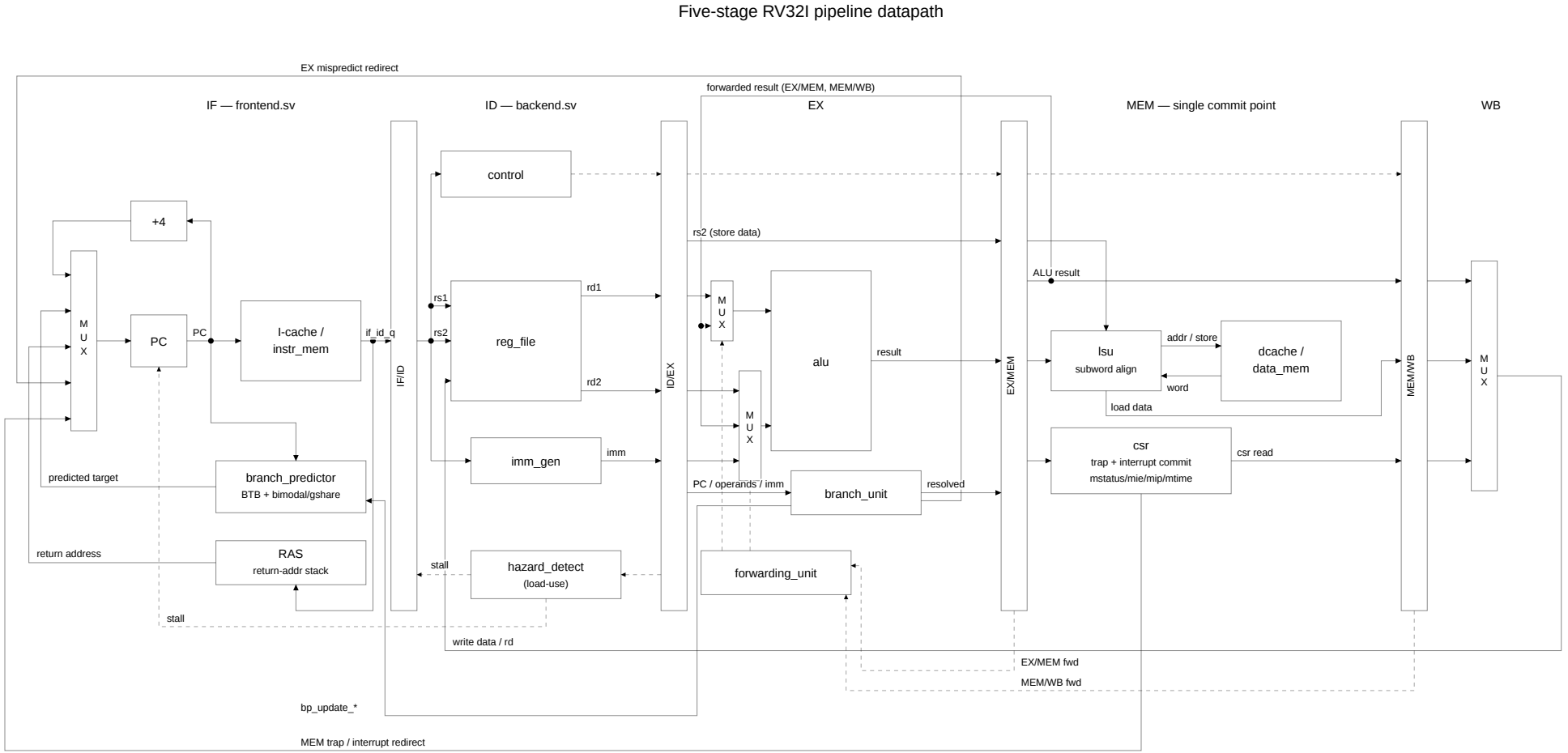
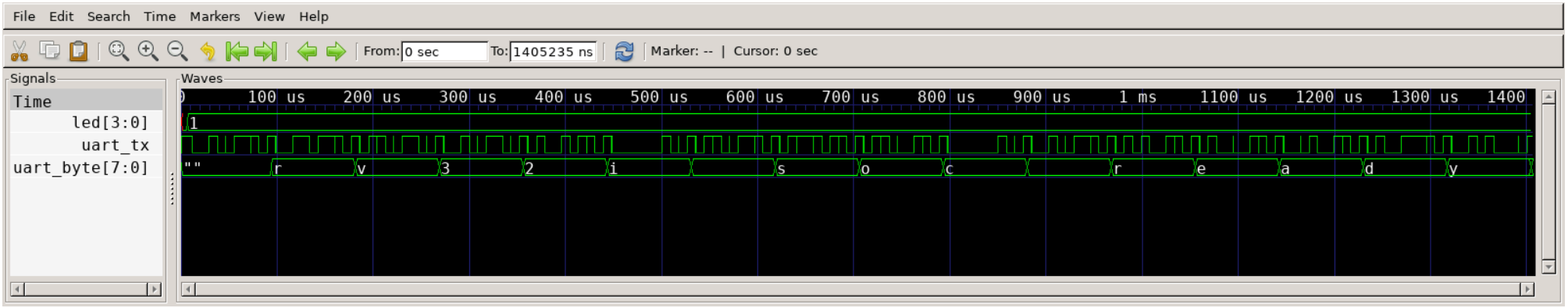
Original project datapath



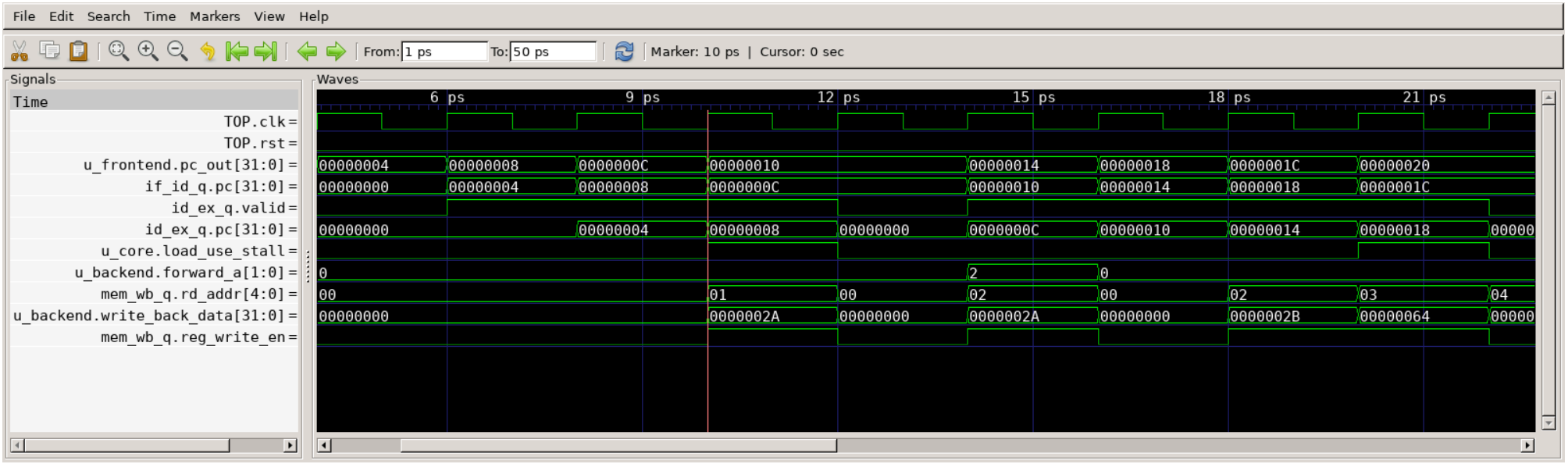
Existing architecture diagram from the repository.

FPGA firmware boot — XSim post-route timing simulation



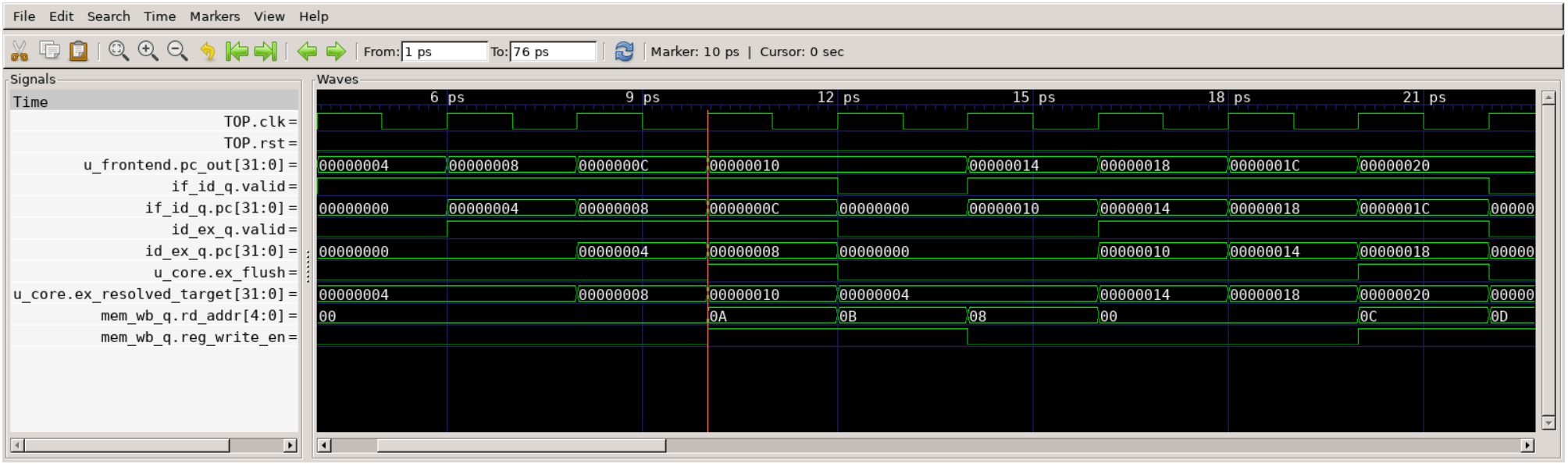
2026-10-20: SDF-annotated FPGA netlist, 50 MHz SoC constraint, +1.847 ns setup slack. UART boot and LED checks passed; DRC warnings remain in the FPGA report.

Load-use hazard



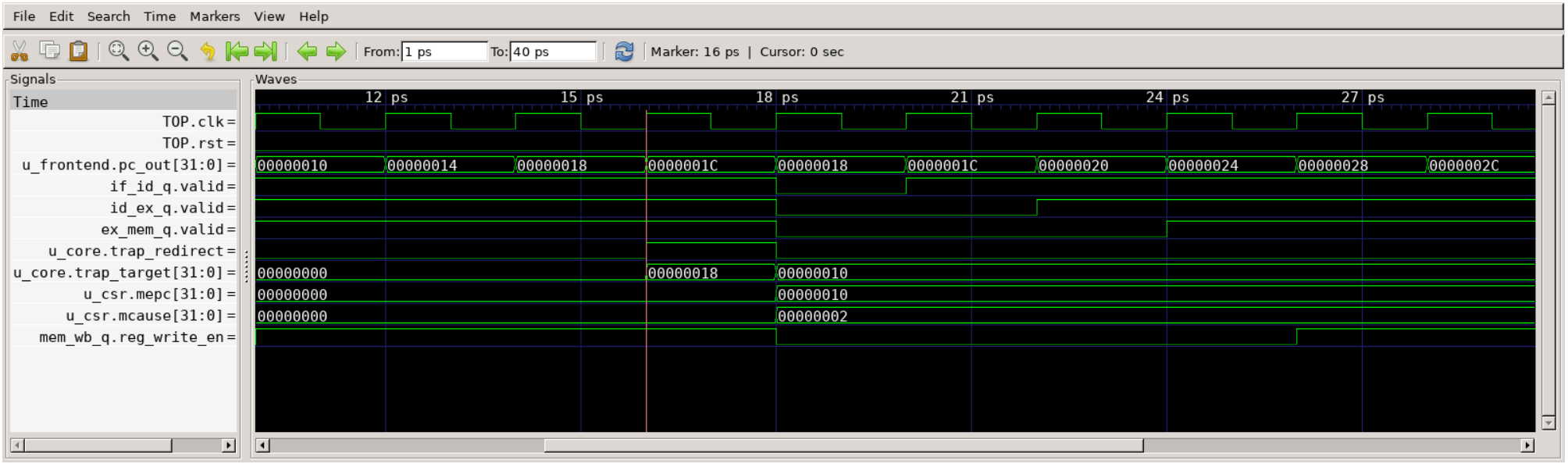
GTKWave screenshot of t07_load_use.vcd; directed test passed.

Branch recovery



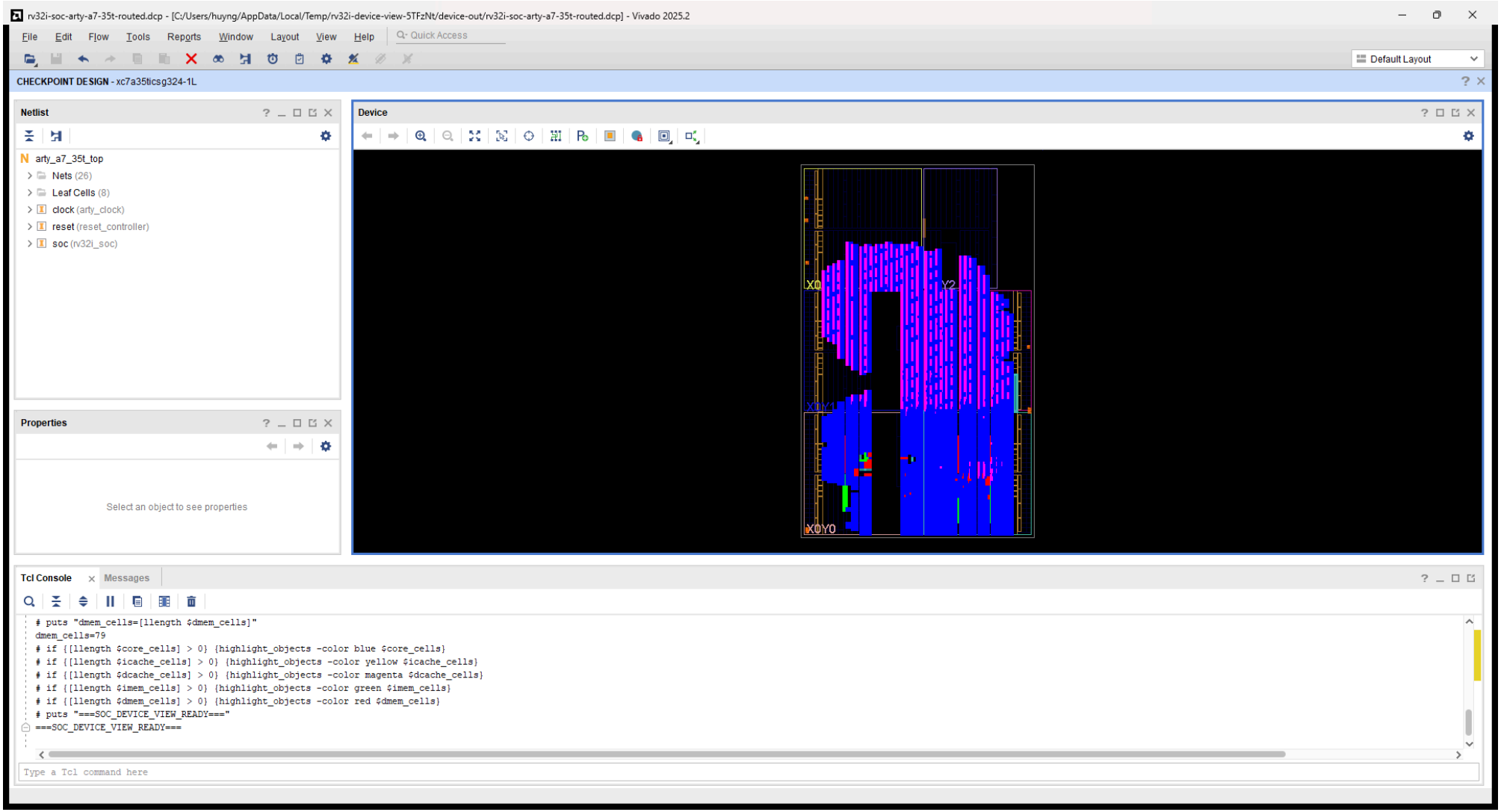
GTKWave screenshot of t04_branch.vcd; directed test passed.

Illegal-instruction trap



GTKWave screenshot of t09_trap_illegal.vcd; directed test passed.

Original Vivado Device view



Historical routed SoC: 50 MHz constraint, +2.009 ns WNS. RTL d21ecd3fe310. No physical-board claim.